Day 30

Serial-In Serial-Out (SISO) Shift Register in Verilog

Circuit Diagram :

Fig 7.9(a)

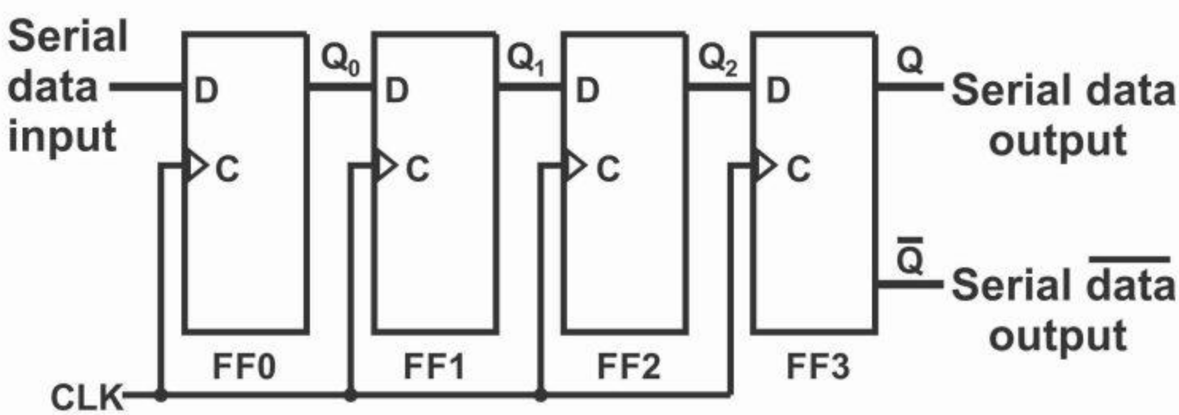


Fig 7.9(b): Serial in/srial out shift register

Verilog Code

4-bit Shift Register verilog code

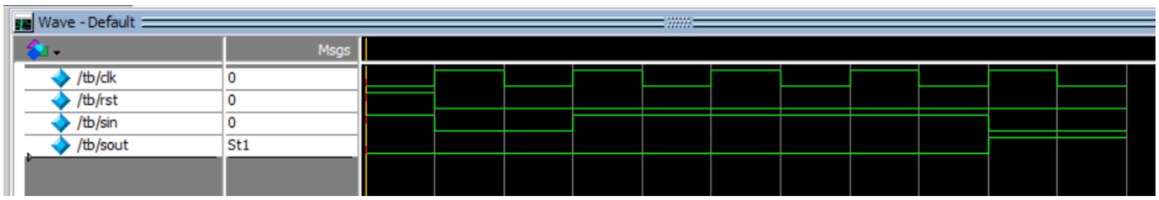
```
module SISO(clk,reset,sin,sout);
input sin,clk,reset;
output sout;
reg [3:0]Q;
always @(posedge clk or posedge reset)
begin
if(reset)
Q<=4'b0000;
else
Q<={Q[2:0],sin};
end
assign sout=Q[3];
endmodule
```

Testbench Code

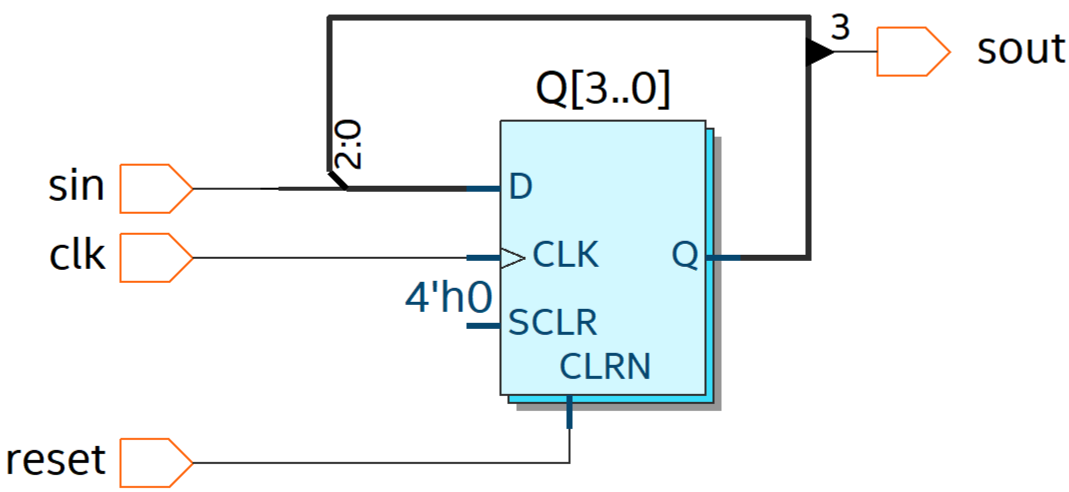
Tb code :

```
module tb;
reg clk,rst,sin;
wire sout;
SISO dut(clk,rst,sin,sout);
initial
begin
clk=0;
forever #5 clk=~clk;
end
initial
begin
rst=1; sin=1; #5
rst=0; sin=0; #10
sin=1; #10
sin=1; #10
sin=1; #10
sin=0; #10
$finish;
end
endmodule
```

Waveform



Schematic



EDA Tools Used

- IntelQuartusPrime
- ModelSim